

Integration & Applications



Basic antiderivatives (the power rule)

1 Find $\int x^4 dx$.

2 Find $\int 3x^2 dx$.

3 Find $\int \sqrt{x} dx$.

4 Find $\int \frac{1}{x^3} dx$.

Antiderivatives of polynomials

5 Find $\int (4x^3 - 6x + 5) dx$.

6 Find $\int (2x + 7) dx$.

7 Find $\int (x^2 - 4x + 3) dx$.

Antiderivatives of trigonometric and exponential functions

8 Find $\int \cos x dx$.

9 Find $\int \sin x dx$.

10 Find $\int e^x dx$.

Evaluating definite integrals

11 Evaluate $\int_0^2 3x^2 dx$.

12 Evaluate $\int_1^3 2x dx$.

13 Evaluate $\int_0^4 x dx$.

14 Evaluate $\int_{-1}^2 3x^2 dx$.

Definite integrals with polynomial integrands

15 Evaluate $\int_0^2 (x^2 + 1) dx$.

16 Evaluate $\int_1^3 (2x^2 - 3x + 1) dx$.

17 Evaluate $\int_0^1 (6x^2 - 4x + 2) dx$.

18 Evaluate $\int_{-2}^1 (x^2 - x) dx$.

Definite integrals of trigonometric and exponential functions

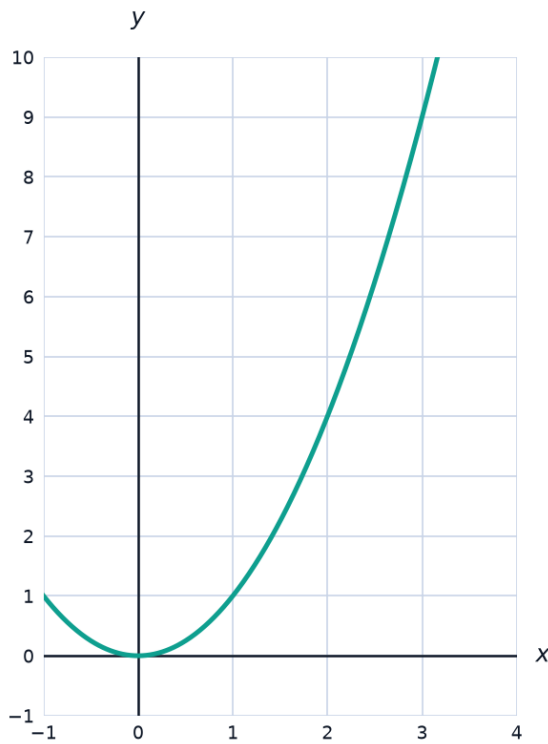
19 Evaluate $\int_0^{\pi/2} \cos x dx$.

20 Evaluate $\int_0^{\pi} \sin x dx$.

21 Evaluate $\int_0^1 e^x dx$.

Area under a curve

22 Find the area under $y = x^2$ from $x = 0$ to $x = 3$, using a definite integral.



23 Find the area under $y = 4 - x^2$ from $x = -2$ to $x = 2$.

- 24 Find the area under $y = 3\sqrt{x}$ from $x = 0$ to $x = 4$.
-

Signed area and regions below the axis

- 25 Evaluate $\int_0^2 (x - 2) dx$ and explain what the sign of the result means.
- 26 Find the (positive) area between $y = x - 2$ and the x -axis, from $x = 0$ to $x = 2$.
-

Properties of definite integrals

- 27 Given $\int_1^4 f(x) dx = 10$, find $\int_1^4 3f(x) dx$.
- 28 Given $\int_0^2 f(x) dx = 5$ and $\int_2^5 f(x) dx = 7$, find $\int_0^5 f(x) dx$.
- 29 Given $\int_0^3 f(x) dx = 8$, find $\int_3^0 f(x) dx$.
-

Finding a function from its derivative

- 30 A function satisfies $f'(x) = 6x^2 - 4$ and $f(1) = 5$. Find $f(x)$.
- 31 A function satisfies $f'(x) = 3x^2 + 2x$ and $f(0) = 1$. Find $f(x)$.
- 32 A function satisfies $f'(x) = 4x - 1$ and $f(2) = 10$. Find $f(x)$.
-

Average value of a function

- 33 Find the average value of $f(x) = x^2$ on $[0, 3]$, using $\text{avg} = \frac{1}{b-a} \int_a^b f(x) dx$.
- 34 Find the average value of $f(x) = 2x + 1$ on $[1, 5]$.
-

Word problems: integration in context

- 35 This question has two parts. A particle moves along a line with velocity $v(t) = 6t - 3t^2$ (m/s) for $0 \leq t \leq 4$.
- a Find the particle's displacement from $t = 0$ to $t = 2$, using $\int_0^2 v(t) dt$.
 - b Find the particle's displacement from $t = 0$ to $t = 4$, and explain why it is not simply twice the part (a) answer.
- 36 This question has two parts. Water flows into a tank at a rate of $r(t) = 20 - 2t$ liters per minute, for $0 \leq t \leq 10$.
- a Write a definite integral for the total volume of water that flows in during the first 5 minutes.
 - b Evaluate the integral to find the volume.
-

MCQ — Basic antiderivatives (the power rule)

- 37 Find $\int x^6 dx$.
- a. $\frac{x^6}{7} + C$
 - b. $7x^5 + C$
 - c. $\frac{x^7}{6} + C$
 - d. $\frac{x^7}{7} + C$
- 38 Find $\int 5x^3 dx$.
- a. $5x^4 + C$
 - b. $\frac{5x^4}{4} + C$
 - c. $\frac{5x^4}{3} + C$
 - d. $15x^2 + C$
- 39 Find $\int \sqrt[3]{x} dx$.
- a. $\frac{3}{2}x^{2/3} + C$
 - b. $\frac{4}{3}x^{4/3} + C$
 - c. $\frac{3}{4}x^{4/3} + C$
 - d. $\frac{1}{3}x^{-2/3} + C$

40 Find $\int \frac{1}{x^4} dx$.

- a. $\frac{1}{3x^3} + C$
- b. $-\frac{1}{4x^3} + C$
- c. $-\frac{1}{3x^3} + C$
- d. $-\frac{4}{x^5} + C$

41 Find $\int 7 dx$.

- a. C
- b. $7x + C$
- c. $7 + C$
- d. $\frac{7x^2}{2} + C$

42 Find $\int \frac{1}{\sqrt{x}} dx$.

- a. $2\sqrt{x} + C$
- b. $\frac{1}{2}\sqrt{x} + C$
- c. $-2\sqrt{x} + C$
- d. $\sqrt{x} + C$

MCQ — Antiderivatives of polynomials

43 Find $\int (6x^2 - 8x + 3) dx$.

- a. $2x^3 - 4x^2 + 3x + C$
- b. $2x^3 - 4x^2 + C$
- c. $2x^3 - 8x^2 + 3x + C$
- d. $12x - 8 + C$

- 44 Find $\int (5x + 9) dx$.
- a. $\frac{5x^2}{2} + C$
 - b. $5x^2 + 9x + C$
 - c. $\frac{5x^2}{2} + 9x + C$
 - d. $\frac{5x^2}{2} + 9 + C$
- 45 Find $\int (x^2 - 6x + 5) dx$.
- a. $\frac{x^3}{3} - 3x^2 + 5x + C$
 - b. $\frac{x^3}{3} - 6x^2 + 5x + C$
 - c. $\frac{x^3}{3} - 3x^2 + C$
 - d. $x^3 - 3x^2 + 5x + C$

- 46 Find $\int (3x^3 - 2x) dx$.
- a. $3x^4 - x^2 + C$
 - b. $\frac{3x^4}{4} - x^2 + C$
 - c. $\frac{3x^4}{4} - 2x^2 + C$
 - d. $\frac{x^4}{4} - x^2 + C$

MCQ — Antiderivatives of trigonometric and exponential functions

- 47 Find $\int 2\cos x dx$.
- a. $2\sin x + C$
 - b. $-2\sin x + C$
 - c. $2\cos x + C$
 - d. $\sin(2x) + C$

- 48 Find $\int 3\sin x \, dx$.
- a. $3\sin x + C$
 - b. $-3\sin x + C$
 - c. $3\cos x + C$
 - d. $-3\cos x + C$
- 49 Find $\int (e^x + 2) \, dx$.
- a. $e^x + 2 + C$
 - b. $e^x + 2x + C$
 - c. $xe^x + 2x + C$
 - d. $e^x + C$
- 50 Find $\int (\sin x + \cos x) \, dx$.
- a. $-\cos x - \sin x + C$
 - b. $\cos x + \sin x + C$
 - c. $\cos x - \sin x + C$
 - d. $-\cos x + \sin x + C$

MCQ — Evaluating definite integrals

- 51 Evaluate $\int_0^3 2x \, dx$.
- a. 9
 - b. 6
 - c. 3
 - d. 18

52 Evaluate $\int_1^4 3x^2 dx$.

a. 21

b. 48

c. 64

d. 63

53 Evaluate $\int_0^5 x dx$.

a. 5

b. 12.5

c. 25

d. 10

54 Evaluate $\int_{-2}^1 4x^3 dx$.

a. -17

b. -15

c. 15

d. 1

55 Evaluate $\int_2^5 3 dx$.

a. 3

b. 9

c. 6

d. 15

56 Evaluate $\int_0^2 6x^2 dx$.

- a. 16
- b. 12
- c. 24
- d. 8

MCQ — Definite integrals with polynomial integrands

57 Evaluate $\int_0^3 (x^2 + 2) dx$.

- a. 9
- b. 15
- c. 11
- d. 21

58 Evaluate $\int_1^4 (3x^2 - 2x + 1) dx$.

- a. 50
- b. 52
- c. 63
- d. 51

59 Evaluate $\int_0^2 (4x^2 - 2x + 3) dx$.

- a. $\frac{32}{3}$
- b. $\frac{40}{3}$
- c. $\frac{38}{3}$
- d. 14

60 Evaluate $\int_{-1}^2 (x^2 - x) dx$.

- a. 4.5
- b. 2.5
- c. 0.5
- d. 1.5

MCQ — Definite integrals of trigonometric and exponential functions

61 Evaluate $\int_0^{\pi/2} \sin x dx$.

- a. $\frac{\pi}{2}$
- b. -1
- c. 0
- d. 1

62 Evaluate $\int_0^{\pi} \cos x dx$.

- a. π
- b. -2
- c. 0
- d. 2

63 Evaluate $\int_0^2 e^x dx$.

- a. $e^2 - 1$
- b. $2e$
- c. $e^2 + 1$
- d. e^2

MCQ — Area under a curve

64 Find the area under $y = x^2$ from $x = 0$ to $x = 4$, using a definite integral.

a. $\frac{64}{3}$

b. 64

c. 16

d. $\frac{16}{3}$

65 Find the area under $y = 9 - x^2$ from $x = -3$ to $x = 3$.

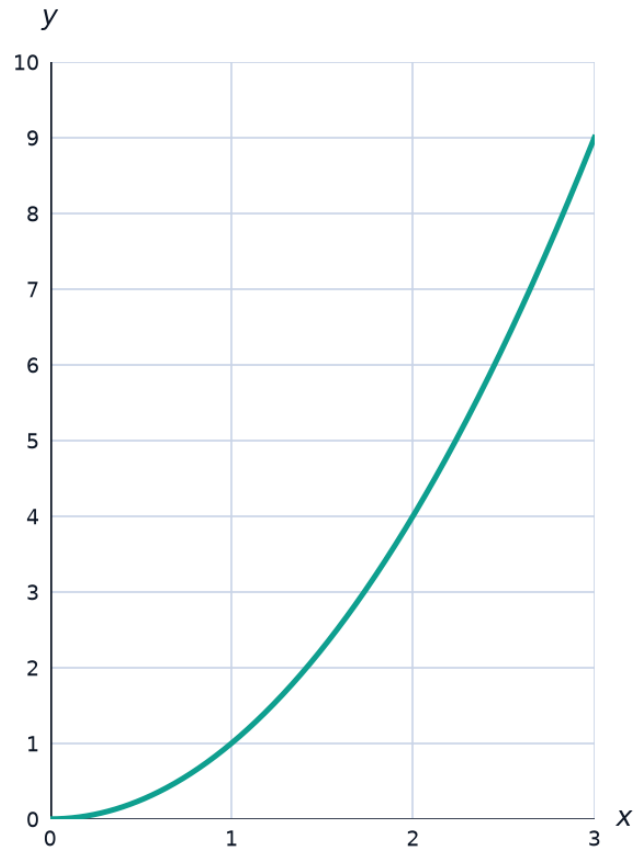
a. 27

b. 54

c. 36

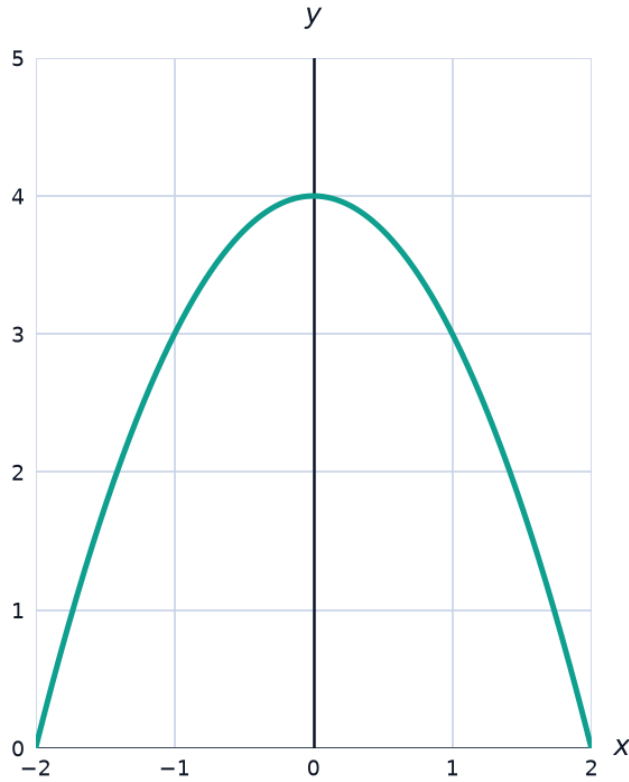
d. 18

- 66 The graph shows $y = x^2$. Find the area under the curve from $x = 0$ to $x = 3$, using a definite integral.



- a. 9
- b. 18
- c. 3
- d. 27

- 67 The graph shows $y = 4 - x^2$. Find the area under the curve from $x = -2$ to $x = 2$.



- a. $\frac{16}{3}$
- b. 32
- c. 16
- d. $\frac{32}{3}$

MCQ — Signed area and regions below the axis

- 68 Evaluate $\int_0^3 (x - 3) dx$ and explain what the sign of the result means.
- a. 4.5; the region lies above the x-axis
 - b. -4.5; the region lies above the x-axis
 - c. 4.5; the region lies below the x-axis
 - d. -4.5; the region lies below the x-axis

- 69 Find the (positive) area between $y = x - 3$ and the x -axis, from $x = 0$ to $x = 3$.
- a. 9
 - b. -4.5
 - c. 3
 - d. 4.5
- 70 Evaluate $\int_0^4 (x - 4) dx$ and state whether the region is above or below the x -axis.
- a. 8; above
 - b. -8 ; above
 - c. 8; below
 - d. -8 ; below
-

MCQ — Properties of definite integrals

- 71 Given $\int_2^6 f(x) dx = 12$, find $\int_2^6 4f(x) dx$.
- a. 12
 - b. 16
 - c. 48
 - d. 3
- 72 Given $\int_0^3 f(x) dx = 6$ and $\int_3^7 f(x) dx = 9$, find $\int_0^7 f(x) dx$.
- a. 13
 - b. 3
 - c. 15
 - d. 54

- 73 Given $\int_0^5 f(x) dx = 14$, find $\int_5^0 f(x) dx$.
- a. 14
 - b. $\frac{1}{14}$
 - c. 0
 - d. -14
- 74 Given $\int_1^5 f(x) dx = 20$ and $\int_1^5 3 dx = 12$, find $\int_1^5 [f(x) - 3] dx$.
- a. -8
 - b. 8
 - c. 32
 - d. 17
- 75 Given $\int_0^4 f(x) dx = 9$ and $\int_0^4 g(x) dx = 5$, find $\int_0^4 [f(x) + g(x)] dx$.
- a. $\frac{9}{5}$
 - b. 45
 - c. 4
 - d. 14

MCQ — Finding a function from its derivative

- 76 A function satisfies $f'(x) = 8x^3 - 6$ and $f(1) = 3$. Find $f(x)$.
- a. $2x^4 - 6x + 7$
 - b. $2x^4 - 6x + 3$
 - c. $8x^4 - 6x + 7$
 - d. $2x^4 - 6x - 7$

77 A function satisfies $f'(x) = 4x^2 + 3x$ and $f(0) = 2$. Find $f(x)$.

- a. $\frac{4x^3}{3} + 3x^2 + 2$
- b. $\frac{4x^3}{3} + 1.5x^2$
- c. $\frac{4x^3}{3} + 1.5x^2 + 2$
- d. $4x^3 + 1.5x^2 + 2$

78 A function satisfies $f'(x) = 6x - 5$ and $f(3) = 20$. Find $f(x)$.

- a. $3x^2 - 5x + 20$
- b. $3x^2 - 5x + 8$
- c. $6x^2 - 5x + 8$
- d. $3x^2 - 5x - 8$

79 A function satisfies $f'(x) = 2x + 7$ and $f(-1) = 4$. Find $f(x)$.

- a. $x^2 + 7x + 10$
- b. $2x^2 + 7x + 10$
- c. $x^2 + 7x - 10$
- d. $x^2 + 7x + 4$

MCQ — Average value of a function

80 Find the average value of $f(x) = x^2$ on $[0, 4]$, using $\text{avg} = \frac{1}{b-a} \int_a^b f(x) dx$.

- a. $\frac{64}{3}$
- b. 16
- c. $\frac{16}{3}$
- d. 4

81 Find the average value of $f(x) = 3x + 2$ on $[0, 4]$.

- a. 32
- b. 4
- c. 14
- d. 8

82 Find the average value of $f(x) = x^2 - 1$ on $[1, 3]$.

- a. $\frac{8}{3}$
- b. $\frac{10}{3}$
- c. $\frac{20}{3}$
- d. 6

MCQ — Word problems: integration in context

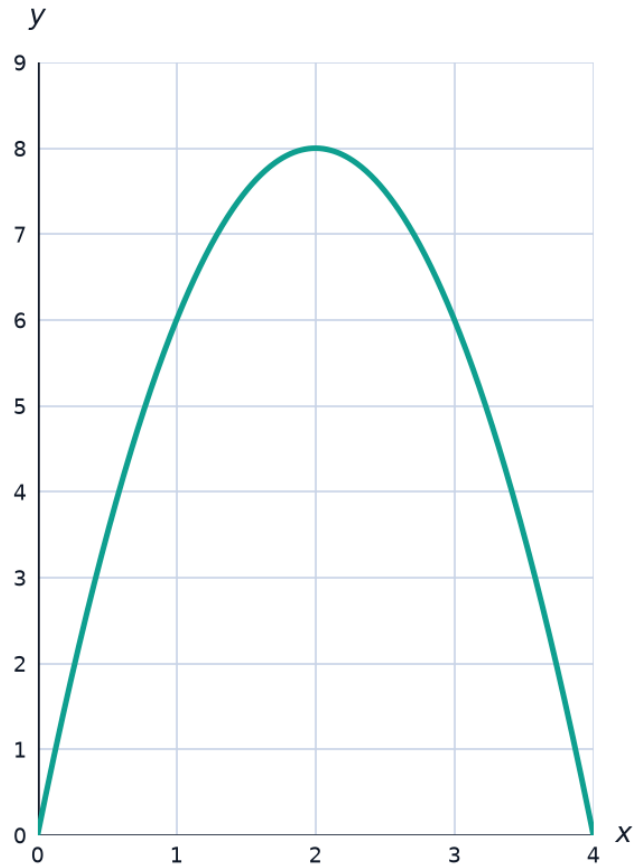
83 A particle moves along a line with velocity $v(t) = 8t - 2t^2$ (m/s) for $0 \leq t \leq 4$. Find the particle's displacement over this interval.

- a. 64
- b. $\frac{64}{3}$
- c. 32
- d. $\frac{192}{3}$

84 Water flows into a tank at a rate of $r(t) = 30 - 3t$ liters per minute, for $0 \leq t \leq 10$. Find the total volume of water that flows in during this time.

- a. 300
- b. 150
- c. 450
- d. 100

- 85 The graph shows a particle's velocity $v(t) = 8t - 2t^2$ for $0 \leq t \leq 4$. Find the particle's displacement (the area under the curve).



- a. $\frac{64}{3}$
- b. 16
- c. 32
- d. 64
- 86 A factory's production rate is $p(t) = 50 + 4t$ items per hour, for $0 \leq t \leq 8$. Find the total number of items produced during this 8-hour period.
- a. 528
- b. 656
- c. 450
- d. 400

Multiple choice — easy

87 Find $\int -x^4 dx$.

a. $-\frac{1}{5}x^5 + C$

b. $-\frac{1}{4}x^4 + C$

c. $-x^5 + C$

d. $-\frac{1}{6}x^5 + C$

88 Find $\int x^4 dx$.

a. $\frac{1}{4}x^4 + C$

b. $x^5 + C$

c. $\frac{1}{6}x^5 + C$

d. $\frac{1}{5}x^5 + C$

89 Find $\int -3x^4 dx$.

a. $-3x^5 + C$

b. $-\frac{3}{5}x^5 + C$

c. $-\frac{1}{2}x^5 + C$

d. $-\frac{3}{4}x^4 + C$

90 Find $\int -3x^5 dx$.

a. $-\frac{3}{5}x^5 + C$

b. $-\frac{1}{2}x^6 + C$

c. $-3x^6 + C$

d. $-\frac{3}{7}x^6 + C$

91 Find $\int -x^2 dx$.

a. $-\frac{1}{3}x^3 + C$

b. $-\frac{1}{4}x^3 + C$

c. $-\frac{1}{2}x^2 + C$

d. $-x^3 + C$

92 Find $\int (-2x + 5) dx$.

a. $-2x + 5 + C$

b. $-x^2 + 7x + C$

c. $-x^2 + 5x + C$

d. $-x^2 - 5x + C$

93 Find $\int (8x + 3) dx$.

a. $4x^2 - 3x + C$

b. $4x^2 + 5x + C$

c. $4x^2 + 3x + C$

d. $8x + 3 + C$

94 Find $\int (-4x - 1) dx$.

a. $-2x^2 + 2x + C$

b. $-2x^2 + x + C$

c. $-2x^2 - x + C$

d. $-4x - 1 + C$

95 Find $\int (4x + 9) dx$.

- a. $2x^2 + 9x + C$
- b. $2x^2 - 9x + C$
- c. $4x + 9 + C$
- d. $2x^2 + 11x + C$

96 Find $\int (-8x - 2) dx$.

- a. $-4x^2 - 2x + C$
- b. $-4x^2 + 2x + C$
- c. $-4x^2 + C$
- d. $-8x - 2 + C$

97 Find $\int \sin x dx$.

- a. $\cos x + C$
- b. $-\cos x + C$
- c. $\sin x + C$
- d. $-\sin x + C$

98 Find $\int 4\cos x dx$.

- a. $4 - \sin x + C$
- b. $4\cos x + C$
- c. $5\sin x + C$
- d. $4\sin x + C$

99 Find $\int 5\cos x \, dx$.

- a. $5 - \sin x + C$
- b. $6\sin x + C$
- c. $5\cos x + C$
- d. $5\sin x + C$

100 Find $\int 3\cos x \, dx$.

- a. $3 - \sin x + C$
- b. $4\sin x + C$
- c. $3\sin x + C$
- d. $3\cos x + C$

101 Find $\int 2\sin x \, dx$.

- a. $-3\cos x + C$
- b. $-2\cos x + C$
- c. $2\sin x + C$
- d. $2\cos x + C$

Multiple choice — medium

102 Evaluate $\int_0^3 3 \, dx$.

- a. -9
- b. 9
- c. 11
- d. 7

103 Evaluate $\int_0^4 4 \, dx$.

- a. -16
- b. 14
- c. 16
- d. 18

104 Evaluate $\int_1^3 6x \, dx$.

- a. 22
- b. 26
- c. 24
- d. -24

105 Evaluate $\int_0^4 2x \, dx$.

- a. 14
- b. 18
- c. -16
- d. 16

106 Evaluate $\int_1^2 1 \, dx$.

- a. 3
- b. 1
- c. 4
- d. -1

107 Evaluate $\int_0^2 (6x^2 + 2) dx$.

a. -20

b. 20

c. 18

d. 22

108 Evaluate $\int_0^3 (9x^2 - 1) dx$.

a. 76

b. 78

c. 80

d. -78

109 Evaluate $\int_0^1 (6x^2 - 2) dx$.

a. 0

b. 3

c. -2

d. 2

110 Evaluate $\int_0^2 (6x^2 + 4) dx$.

a. 26

b. 24

c. 22

d. -24

111 Evaluate $\int_0^2 (6x^2 - 3) dx$.

- a. 10
- b. 12
- c. -10
- d. 8

112 A function satisfies $f'(x) = -2x + 5$ and $f(1) = 10$. Find $f(x)$.

- a. $f(x) = x^2 + 5x + 6$
- b. $f(x) = -x^2 + 5x + 9$
- c. $f(x) = -x^2 + 5x - 6$
- d. $f(x) = -x^2 + 5x + 6$

113 A function satisfies $f'(x) = 6x + 3$ and $f(2) = -5$. Find $f(x)$.

- a. $f(x) = 3x^2 + 3x - 23$
- b. $f(x) = 3x^2 + 3x - 20$
- c. $f(x) = -3x^2 + 3x - 23$
- d. $f(x) = 3x^2 + 3x + 23$

114 A function satisfies $f'(x) = 2x + 6$ and $f(1) = -6$. Find $f(x)$.

- a. $f(x) = x^2 + 6x + 13$
- b. $f(x) = x^2 + 6x - 10$
- c. $f(x) = x^2 + 6x - 13$
- d. $f(x) = -x^2 + 6x - 13$

115 A function satisfies $f'(x) = -2x + 2$ and $f(0) = 4$. Find $f(x)$.

a. $f(x) = -x^2 + 2x - 4$

b. $f(x) = x^2 + 2x + 4$

c. $f(x) = -x^2 + 2x + 7$

d. $f(x) = -x^2 + 2x + 4$

116 A function satisfies $f'(x) = -6x$ and $f(1) = -6$. Find $f(x)$.

a. $f(x) = 3x^2 - 3$

b. $f(x) = -3x^2 - 3$

c. $f(x) = -3x^2$

d. $f(x) = -3x^2 + 3$

Multiple choice — hard

117 Find the area under $y = 3x^2$ from $x = 0$ to $x = 4$, using a definite integral.

a. 192

b. 66

c. 62

d. 64

118 Find the area under $y = 3x^2$ from $x = 0$ to $x = 2$, using a definite integral.

a. 10

b. 8

c. 6

d. 24

119 Find the area under $y = 2x^2$ from $x = 0$ to $x = 3$, using a definite integral.

- a. 18
- b. 20
- c. 54
- d. 16

120 Find the area under $y = x^2$ from $x = 0$ to $x = 3$, using a definite integral.

- a. 7
- b. 27
- c. 9
- d. 11

121 Find the area under $y = 3x^2$ from $x = 0$ to $x = 3$, using a definite integral.

- a. 25
- b. 81
- c. 27
- d. 29

122 Given $\int_0^2 f(x) dx = 5$ and $\int_2^6 f(x) dx = 2$, find $\int_0^6 f(x) dx$.

- a. 3
- b. 9
- c. 10
- d. 7

123 Given $\int_0^5 f(x) dx = 10$, find $\int_5^0 f(x) dx$.

- a. -10
- b. 12
- c. -12
- d. 10

124 Given $\int_1^4 f(x) dx = 7$, find $\int_1^4 3f(x) dx$.

- a. -21
- b. 10
- c. 21
- d. 23

125 Given $\int_0^2 f(x) dx = 6$ and $\int_2^6 f(x) dx = 5$, find $\int_0^6 f(x) dx$.

- a. 11
- b. 13
- c. 30
- d. 1

126 Given $\int_0^5 f(x) dx = 3$, find $\int_5^0 f(x) dx$.

- a. 3
- b. -3
- c. -5
- d. 5

127 Find the average value of $f(x) = x + 1$ on $[0, 4]$.

- a. 5
- b. 2
- c. 4
- d. 3

128 Find the average value of $f(x) = 3x - 3$ on $[0, 2]$.

- a. 1
- b. 2
- c. -1
- d. 0

129 Find the average value of $f(x) = 2x - 4$ on $[1, 5]$.

- a. 3
- b. 1
- c. 4
- d. 2

130 Find the average value of $f(x) = -2x + 4$ on $[1, 4]$.

- a. -1
- b. -2
- c. 0
- d. 1

131 Find the average value of $f(x) = 2x + 3$ on $[1, 3]$.

a. 9

b. 8

c. 7

d. 6